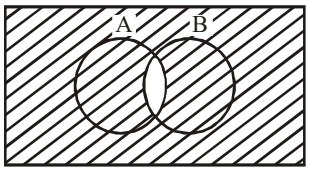
SOLUTION

SECTION–I

1. Ans. (A)



$= \cup - (A \cap B)$

$\Rightarrow \overline{(A \cap B)} = \bar{A} \cup \bar{B}$

2. Ans. (C)

Sol. $n_1 = 100 \quad m = 250$

$\bar{X}_1 = 15 \quad \bar{X} = 15.6$

$V_1(x) = 9 \quad Var(x) = 13.44$

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

$n_2 = 150, \quad \bar{x}_2 = 16, \; V_2(x) = \sigma_2$

$$13.44 = \frac{100 \times 9 + 150 \times \sigma_2^2}{250} + \frac{100 \times 150}{(250)^2} \times 1$$

$\Rightarrow \sigma_2^2 = 16 \Rightarrow \sigma_2 = 4$

3. Ans. (D)

Sol. Given :

$$\text{Mean } (\bar{x}) = \frac{\Sigma x_i}{20} = 10$$

or $\Sigma x_i = 200$ (incorrect)

or $200 - 25 + 35 = 210 = \Sigma x_i$ (Correct)

$$\text{Now correct } \bar{x} = \frac{210}{20} = 10.5$$

again given S.D = 2.5 (σ)

$$\sigma^2 = \frac{\Sigma x_i^2}{20} - (10)^2 = (2.5)^2$$

or $\Sigma x_i^2 = 2125$ (incorrect)

or $\Sigma x_i^2 = 2125 - 25^2 + 35^2$

$= 2725$ (Correct)

$$\therefore \text{ correct } \sigma^2 = \frac{2725}{20} - (10.5)^2$$

$\sigma^2 = 26$

or $\sigma = 26$

$\therefore \alpha = 10.5, \; \beta = 26$

4. Ans. (B)

$2^m = 56 + 2^n$

$\Rightarrow 2^m - 2^n = 8 \times 7$

$\Rightarrow 2^{m-3} - 2^{n-3} = 7$

even – even cant be odd

$\therefore 2^{n-3} = \text{odd} = 1$, if $n = 3$

$m = 6$

$m + n = 9.$

5. Ans. (D)

Sol. For a, b, c

$$\text{mean} = \frac{a+b+c}{3} (= \bar{x})$$

$b = a + c$

$$\Rightarrow \bar{x} = \frac{2b}{3} \qquad .....(1)$$

S.D. (a + 2, b + 2, c + 2) = S.D. (a, b, c) = d

$$\Rightarrow \quad d^2 = \frac{a^2 + b^2 + c^2}{3} - (\bar{x})^2$$

$$\Rightarrow \quad d^2 = \frac{a^2 + b^2 + c^2}{3} - \frac{4b^2}{9}$$

$\Rightarrow 9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$

$\Rightarrow b^2 = 3(a^2 + c^2) - 9d^2$

6. Ans. (A)

Sol. Let observations are denoted by xi for $1 \le i < 2n$

$$\bar{x} = \frac{\sum x_i}{2n} = \frac{(a + a + ... + a) - (a + a + ... + a)}{2n}$$

$\Rightarrow \bar{x} = 0$

$$\text{and } \sigma_x^2 = \frac{\sum x_i^2}{2n} - (\bar{x})^2 = \frac{a^2 + a^2 + ... + a^2}{2n} - 0 = a^2$$

$\Rightarrow \sigma_x = a$

Now, adding a constant b then $\bar{y} = \bar{x} + b = 5$

$\Rightarrow b = 5$

and $\sigma_y = \sigma_x$ (No change in S.D.) $\Rightarrow a = 20$

$\Rightarrow a^2 + b^2 = 425$

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7. Ans. (B)

Reflexive : a + a = 2a need not to be irrational
$\Rightarrow$ not reflexive

Symmetric :
If a + b = irrational number , then
b + a = irrational number
$\therefore$ M is symmetric relation

Transistive :

$(1,\sqrt{2}) \in M$

$(\sqrt{2},2) \in M$

but $(1,2) \notin M$
$\therefore$ It is not transistive

8. Ans. (B)

xRx as sinx = sinx $\quad\therefore$ Reflexive

xRy $\Rightarrow$ yRx $\quad\therefore$ Symmetric

xRy & yRz $\Rightarrow$ xRz $\quad\therefore$ transitive

$\Rightarrow$ Relation is equivalence

9. Ans. (C)

$|a| \le b$ and $|b| \le c$

$\Rightarrow \quad |a| \le c$

$\Rightarrow$ relation is transitive

$|-1| \le 5$ but $|5| \not\le -1$

$\Rightarrow$ relation is not symmetric

$|-2| \not\le -2$

$\Rightarrow$ relation is not reflexive

10. Ans. (D)

If variance of $\sum x_i$ is v

then variance of $\sum ax_i + b$ will be a^2v

11. Ans. (D)

Mean is changed as the observations are changed. Median will remain same.

12. Ans. (B)

$x_1 + x_2 + x_3 + \ldots\ldots + x_n = \bar{x}.n$

$$\Rightarrow \quad \frac{x_1}{\alpha}+\frac{x_2}{\alpha}+\frac{x_3}{\alpha}\ldots\ldots+\frac{x_n}{\alpha}=\frac{\bar{x}.n}{\alpha}$$

$$\Rightarrow \quad \frac{x_1}{\alpha}+\beta+\frac{x_2}{\alpha}+\beta+\ldots\ldots+\frac{x_n}{\alpha}+\beta=\frac{\bar{x}.n}{\alpha}+n\beta$$

$$\Rightarrow \quad \frac{\frac{x_1}{\alpha}+\beta+\frac{x_2}{\alpha}+\beta+\ldots..+\frac{x_n}{\alpha}+\beta}{n}=\frac{\bar{x}}{\alpha}+\beta$$

13. Ans. (B)

Mean square deviation is minimum about mean i.e. a_{51} and median of the group is a_{51}
$\Rightarrow$ Mode will also be a_{51}

14. Ans. (A)

Median is (5.5x) = a

$$\text{Mean deviation} = \frac{\sum|x_i - a|}{10} = 30$$

$$\frac{2(4.5x+3.5x+2.5x+1.5x+.5x)}{10} = 30$$

$$\frac{2(12.5)|x|}{10} = 30$$

$|x| = 12$

15. Ans. (B)

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1+n_2} + \frac{n_1n_2}{(n_1+n_2)^2}\ (\bar{x}_1 - \bar{x}_2)^2$$

$$\sigma^2 = \frac{6\times 24 + 3\times 36}{6+3} + \frac{6\times 3}{(6+3)^2}\ (11-14)^2$$

$$\sigma^2 = \frac{144+108}{9} + \frac{18}{81}\times 9 = 28 + 2 = 30$$

16. Ans. (B)

$$\bar{x} = \frac{\sum x_i}{100} = 0 \ \&$$

$$\frac{\sum|x_i - \bar{x}|}{100} = 5 \Rightarrow \sum|x_i| = 500$$

$$\Rightarrow \sum x_i^2 + 2\sum_{1\le i<j\le 100}|x_ix_j| = (500)^2$$

$$\Rightarrow \frac{\sum x_i^2}{100} = \frac{(500)^2 - 2\sum|x_ix_j|}{100} = 2500 - 1600$$

$$\text{S.D.} = \sqrt{\frac{\sum(x_i-\bar{x})^2}{100}} = \sqrt{900} = 30$$

17. Ans. (C)

$$\frac{\Sigma a}{n} = 38 \Rightarrow \Sigma a = (38)(50)$$

$$\frac{\Sigma a - 100}{48} = \frac{(38)(50)-100}{48} = 37.5$$

18. Ans. (D)

Sol. $\frac{\Sigma x_i^2}{n} \ge \left(\frac{\Sigma x_i}{n}\right)^2 \Rightarrow \frac{200}{n} \ge \frac{(40)^2}{n^2} \Rightarrow n > 8$

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19. Ans. (C)

Let x_1, x_2, x_n be n observations, then

$$\bar{X} = \frac{1}{n}\sum x_i$$

Let $y_i = \frac{x_i}{\alpha} + 10$, then

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{\alpha}\left(\frac{1}{n}\sum x_i\right) + \frac{1}{n}(10n)$$

$$\Rightarrow \bar{Y} = \frac{1}{\alpha}\bar{X} + 10 = \frac{\bar{X} + 10\alpha}{\alpha}$$

SECTION–II

1. Ans. 1.00

After adding (C,C) the R is reflexive symmetric and transitive

2. Ans. 12.00

Sol. $5 = \frac{3+7+x+y}{4} \Rightarrow x + y = 10$

$$\text{Var}(x) = 10 = \frac{3^2 + 7^2 + x^2 + y^2}{4} - 25$$

$140 = 49 + 9 + x^2 + y^2$

$x^2 + y^2 = 82$

$x + y = 10$

$\Rightarrow (x,y) = (9,1)$

Four numbers are 21,9,10,8

$$\text{Mean} = \frac{48}{4} = 12$$

3. Ans. 4.00

Sol. $\sum_{i=1}^{18}(X_i - \alpha) = 36$, $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$

$$\Rightarrow \sum_{i=1}^{18} x_i = 18(\alpha + 2),$$

$$\sum_{i=1}^{18} x_i^2 - 2\beta\sum_{i=1}^{18} x_i = 18\beta^2 = 90$$

Hence $\sum x_i^2 = 90 - 18\beta^2 + 36\beta(\alpha + 2)$

Given $\frac{\sum x_i^2}{18} - \left(\frac{\sum x_i}{18}\right)^2 = 1$

$\Rightarrow 90 - 18\beta^2 + 36\beta(\alpha + 2) - 18(\alpha + 2)^2 = 18$

$\Rightarrow 5 - \beta^2 + 2\alpha\beta + 4\beta - \alpha^2 - 4\alpha - 4 = 1$

$\Rightarrow (\alpha - \beta)^2 + 4(\alpha - \beta) = 0 \Rightarrow |\alpha - \beta| = 0$ or 4

As α and β are distinct $|\alpha - \beta| = 4$

4. Ans. 8.00

Var $(ax_i + b) = a^2$ var (x_i)

Variance on doubling each observation

$= 2^2 \times 16 = 64$

Std. deviation $= \sqrt{\text{var}} = 8$

5. Ans. 24.00

Mode = 3 median – 2 mean

$= 66 - 42 = 24$

6. Ans. 1.25

$$\text{M.D.} = \frac{\sum|x_i - 3|}{4} = 1$$

$\Rightarrow 2 + 1 + |x - 3| = 4 \Rightarrow x = 4$

$$\text{Variance} = \frac{\sum(x_i - \bar{x})^2}{4} = \frac{5}{4}$$

where $\bar{x} = \frac{1+2+3+4}{4} = \frac{5}{2}$

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